

Source Reconstruction of EEG data

Using MNE Python

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Outline

Overview Source Reconstruction

Anatomical Scans, Coregistration and BEM

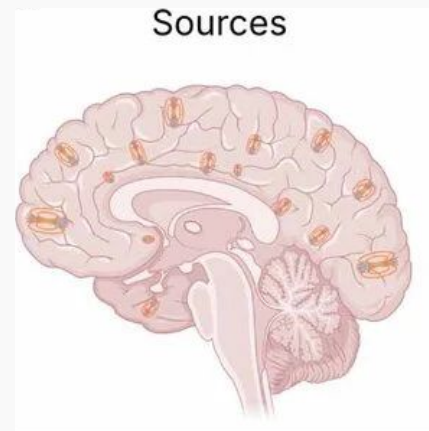
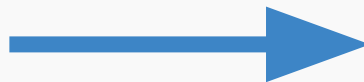
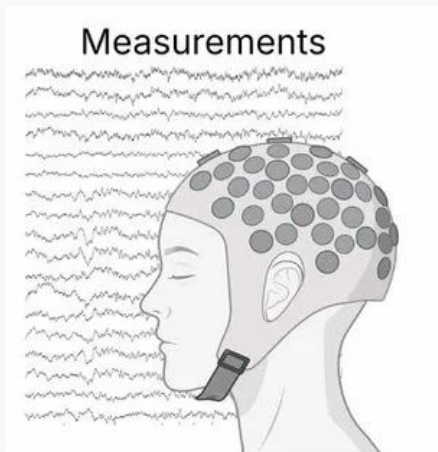
Forward & Inverse Model

Comparing Inverse Models

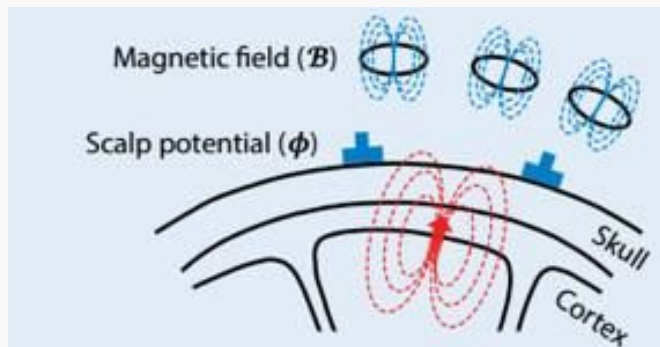
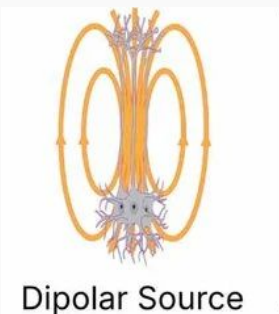
EEG Experiment for Today

What is Source Reconstruction?

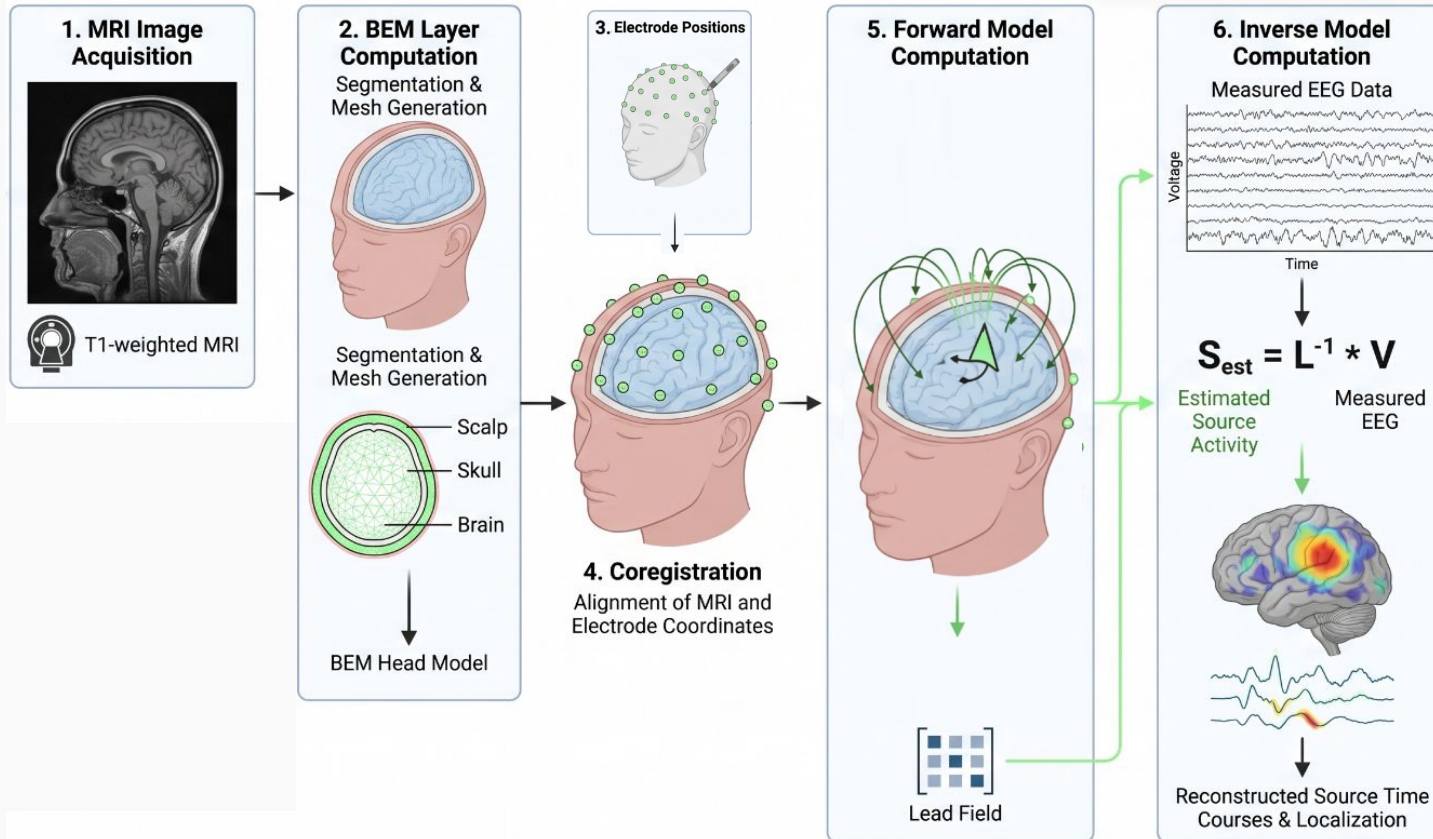
Goal:



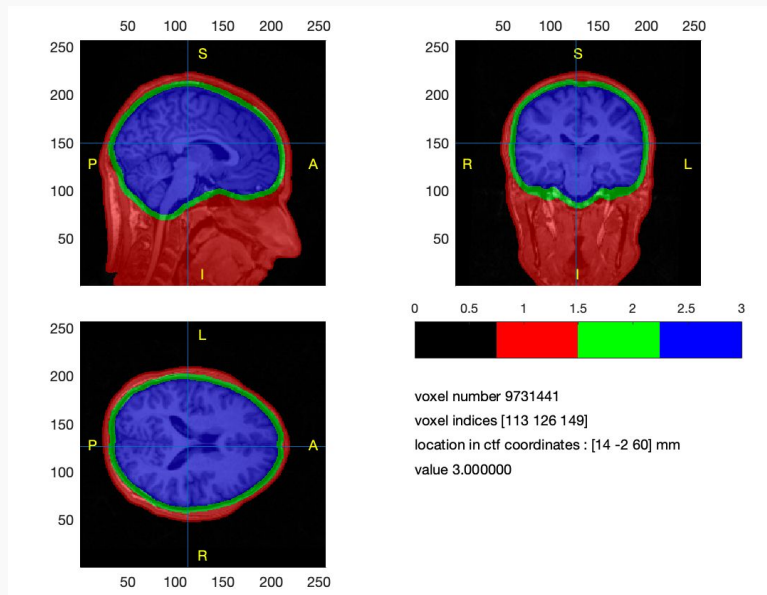
- Generated by firing of pyramidal cell populations
- Two separated electrical charges (apical vs basal dendrite)



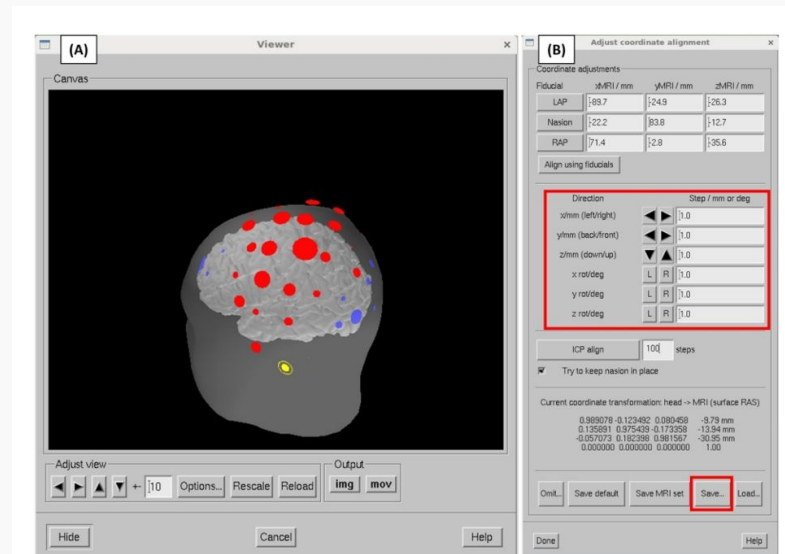
Source Reconstruction Pipeline



Boundary Element Method (BEM)

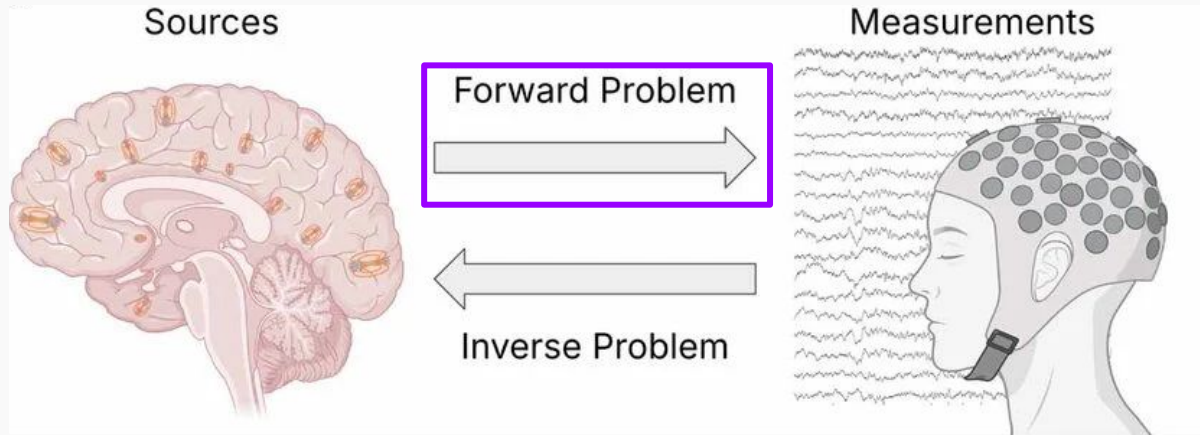


Coregistration

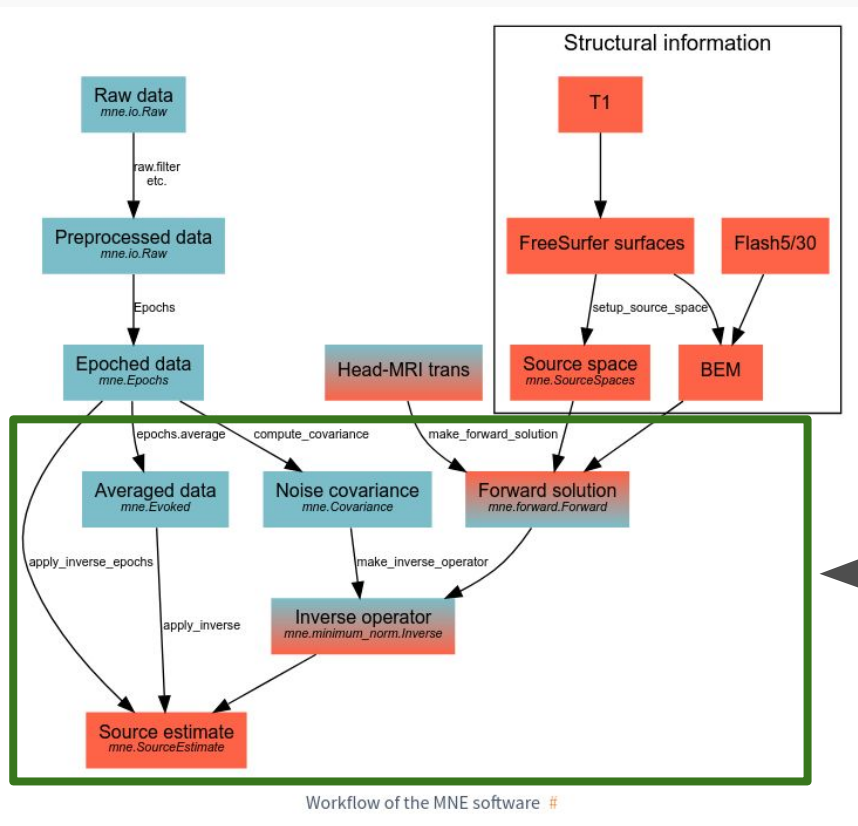


The Forward Model (or Leadfield)

How would a sensor see a source if it were active?



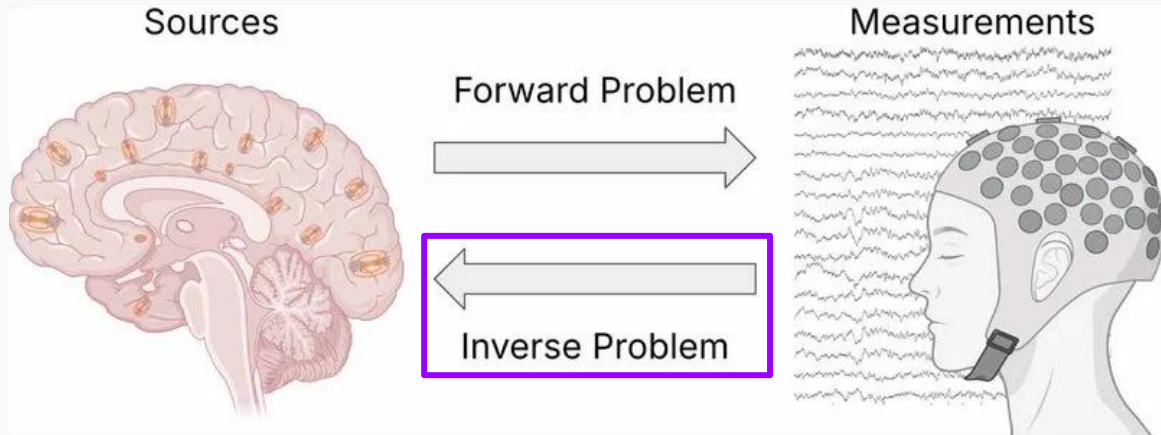
MNE Python Source Reconstruction Workflow



← What we focus on today!

The Inverse Model

What is the source activity given our sensor recordings?



Problem:

- ~**20000** vertices and **62** channels
→ ill-defined problem
- Different configurations could lead to the exact same sensor data that we see in our measured data!

The Inverse Model

What is the source activity given our sensor recordings?

$$\mathbf{y}(t) = \mathbf{L}\mathbf{x}(t) + \mathbf{n}(t)$$

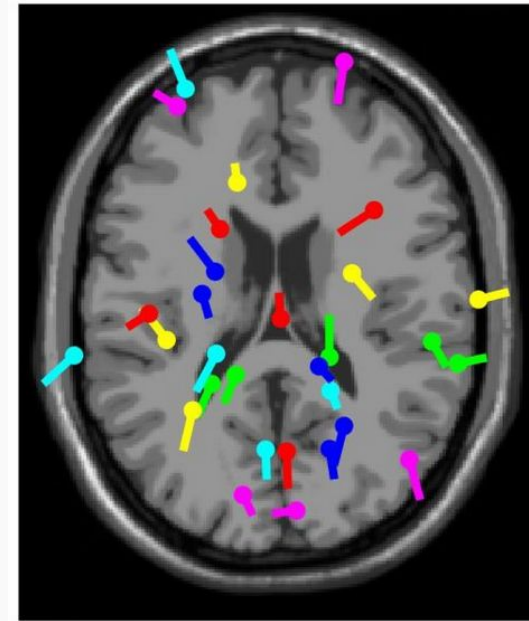
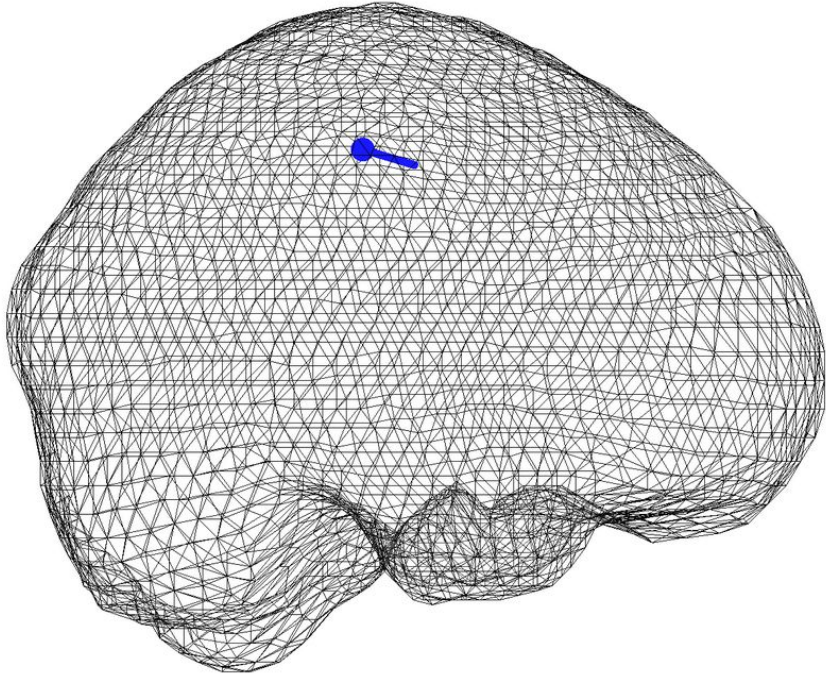


$$\hat{\mathbf{x}}(t) = \mathbf{W}\mathbf{y}(t)$$

symbol	shape	meaning
$\mathbf{y}(t)$	$N \times 1$	measured sensor data at time t - N channels
$\mathbf{L}$	$N \times M$	lead field (or gain matrix) - M possible neural sources
$\mathbf{x}(t)$	$M \times 1$	source amplitudes we want to recover
$\mathbf{n}(t)$	$N \times 1$	noise

The Inverse Model

Assumption we need to make: How many dipoles do we expect?



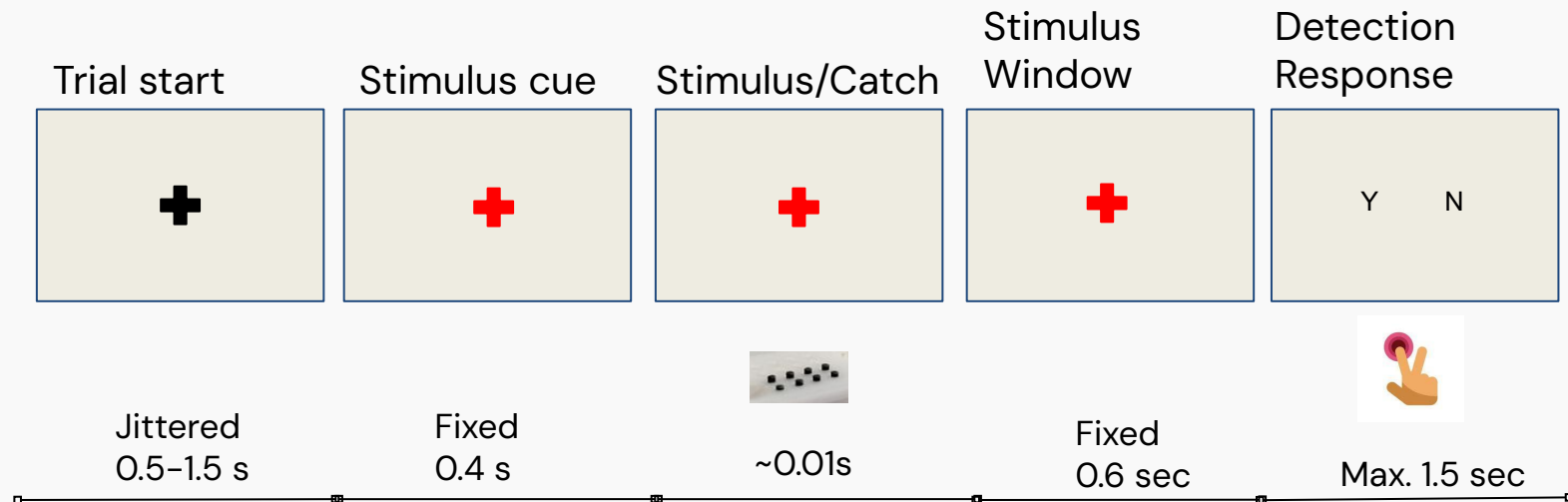
“Distributed Dipole Style” Inverse Models

- Minimum–Norm Estimate
 - Distributed dipole model
 - Each vertex represents a dipole
- Beamformer
 - linear–constrained minimum–variance (LCMV)
- eLAURETA/sLAURETA
- dSPM

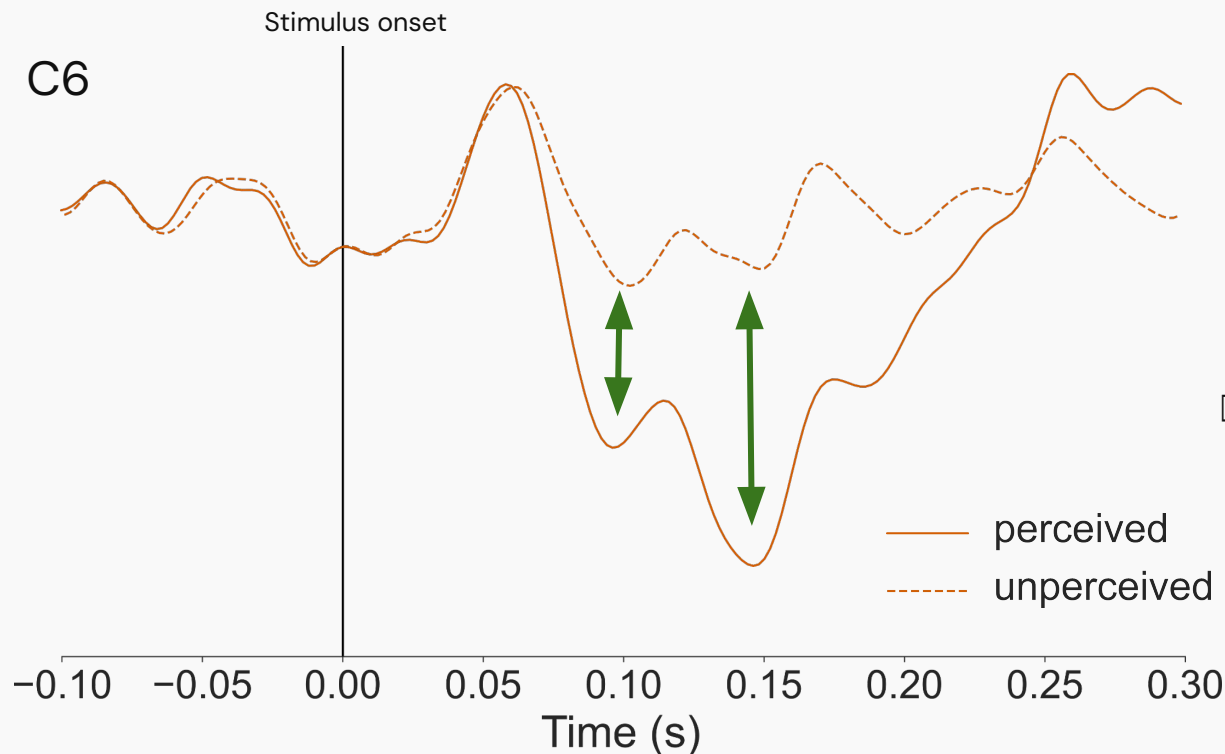
“Distributed Dipole Style” Inverse Models

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EEG Experiment: Somatosensory Near-Threshold Detection



Near-threshold somatosensory perception



- 36 participants
- N140** as reliable neural correlate of reportable perception
- Earlier correlates might be possible!

Let's get Hands-On!

<https://github.com/rosagross/cuttingeeg-2026-sourcerec-tutorial>

Extra slide:

What is the unit of a source reconstructed signal?

This depends on the inverse model (and its options)!

MNE: ampere-metres (Am) $\rightarrow$ electrical current

LCMV: unit-gain: ampere-metres (Am), unit-noise-gain: scales estimated dipole by the standard deviation of the weights applied to the noise covariance, so Am/Am $\rightarrow$ unit-less!